



University of Engineering & Technology Peshawar

Computer Programming (C++) Lab

Assginment 02

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SUBMITTED TO ENGR RIZWAN ULLAH

Q1 – Array Statistics (Max, Min, Average)

Description: Takes 10 integer inputs in an array and finds the maximum value, minimum value, and average.

Source Code (C++):

```
#include <iostream>
using namespace std;

int main() {
    int arr[10], sum = 0, max, min;

    cout << "Enter 10 numbers: ";
    for (int i = 0; i < 10; i++) {
        cin >> arr[i];
    }

    max = min = arr[0];

    for (int i = 0; i < 10; i++) {
        if (arr[i] > max) max = arr[i];
        if (arr[i] < min) min = arr[i];
        sum += arr[i];
    }

    cout << "Maximum = " << max << endl;
    cout << "Minimum = " << min << endl;
    cout << "Average = " << sum / 10.0 << endl;

    return 0;
}
```

Q2 – Reverse a 1D Array Using a Function

Description: Uses a function to reverse a 1D array and displays the array before and after reversal.

Source Code (C++):

```
#include <iostream>
using namespace std;

int main() {
    int arr[10], sum = 0;

    cout << "Enter 10 numbers:\n";
    for(int i = 0; i < 10; i++) {
        cin >> arr[i];
        sum += arr[i];
    }

    int max = arr[0], min = arr[0];

    for(int i = 1; i < 10; i++) {
        if(arr[i] > max) max = arr[i];
        if(arr[i] < min) min = arr[i];
    }

    cout << "Max: " << max << endl;
    cout << "Min: " << min << endl;
    cout << "Average: " << (float)sum / 10 << endl;

    return 0;
}
```

Q3 – 3×3 Matrix Addition

Description: Inputs two 3×3 matrices, performs matrix addition, and displays the result matrix.

Source Code (C++):

```
#include <iostream>
using namespace std;

int main() {
    int A[3][3], B[3][3], C[3][3];

    cout << "Enter Matrix A:\n";
    for(int i=0;i<3;i++)
        for(int j=0;j<3;j++)
            cin >> A[i][j];

    cout << "Enter Matrix B:\n";
    for(int i=0;i<3;i++)
        for(int j=0;j<3;j++)
            cin >> B[i][j];

    for(int i=0;i<3;i++)
        for(int j=0;j<3;j++)
            C[i][j] = A[i][j] + B[i][j];

    cout << "Result Matrix:\n";
    for(int i=0;i<3;i++) {
        for(int j=0;j<3;j++)
            cout << C[i][j] << " ";
        cout << endl;
    }

    return 0;
}
```

Q4 – Alphabetical Sorting of Student Names

Description: Stores names of 5 students in a string array, sorts them alphabetically using bubble sort, and displays the sorted list.

Source Code (C++):

```
#include <iostream>
#include <string>
using namespace std;

int main() {
    string names[5];

    cout << "Enter 5 names:\n";
    for(int i=0;i<5;i++) cin >> names[i];

    for(int i=0;i<5;i++) {
        for(int j=i+1;j<5;j++) {
            if(names[i] > names[j]) {
                string temp = names[i];
                names[i] = names[j];
                names[j] = temp;
            }
        }
    }

    cout << "Sorted Names:\n";
    for(int i=0;i<5;i++) cout << names[i] << endl;

    return 0;
}
```

Q5 – Power System Load Analysis (1D Array + Functions)

Description: Stores hourly load demand (MW) for 24 hours. Functions calculate total load, peak load, and average load.

Source Code (C++):

```
#include <iostream>
using namespace std;

float totalLoad(float arr[]) {
    float sum = 0;
    for(int i=0;i<24;i++) sum += arr[i];
    return sum;
}

float peakLoad(float arr[]) {
    float max = arr[0];
    for(int i=1;i<24;i++)
        if(arr[i] > max) max = arr[i];
    return max;
}

int main() {
    float load[24];

    cout << "Enter 24 hourly loads:\n";
    for(int i=0;i<24;i++) cin >> load[i];

    cout << "Total Load: " << totalLoad(load) << endl;
    cout << "Peak Load: " << peakLoad(load) << endl;

    return 0;
}
```

Q6 – Student Record System (Names, Marks, Search)

Description: Stores student names and marks. Functions find the topper, display pass/fail status (pass mark = 50), and search a student by name.

Source Code (C++):

```
#include <iostream>
#include <string>
using namespace std;

int main() {
    string names[5];
    int marks[5];

    for(int i=0;i<5;i++) {
        cout << "Enter name and marks: ";
        cin >> names[i] >> marks[i];
    }

    // topper
    int max = marks[0], index = 0;
    for(int i=1;i<5;i++) {
        if(marks[i] > max) {
            max = marks[i];
            index = i;
        }
    }

    cout << "Topper: " << names[index] << endl;

    // pass/fail
    for(int i=0;i<5;i++) {
        if(marks[i] >= 50)
            cout << names[i] << " Pass\n";
        else
            cout << names[i] << " Fail\n";
    }

    // search
    string search;
    cout << "Enter name to search: ";
    cin >> search;

    for(int i=0;i<5;i++) {
        if(names[i] == search) {
            cout << "Found with marks: " << marks[i];
        }
    }

    return 0;
}
```

Q7 – Renewable Energy Monitoring System (Solar Plant)

Description: Stores solar energy generation (kWh) for 24 hours. Functions calculate total energy, the hour with maximum generation, and average hourly output.

Source Code (C++):

```
#include <iostream>
using namespace std;

int main() {
    float energy[24], sum = 0, max;

    for(int i=0;i<24;i++) cin >> energy[i];

    max = energy[0];
    int hour = 0;

    for(int i=0;i<24;i++) {
        sum += energy[i];
        if(energy[i] > max) {
            max = energy[i];
            hour = i;
        }
    }

    cout << "Total: " << sum << endl;
    cout << "Max at hour: " << hour << endl;
    cout << "Average: " << sum/24 << endl;

    return 0;
}
```


Q8 – Battery Health Monitoring System

Description: Stores 20 voltage readings. Safe limits: 3.0V – 4.2V. Functions count unsafe readings, find max/min voltage, and display warnings if unsafe readings exist.

Source Code (C++):

```
#include <iostream>
using namespace std;

int main() {
    float v[20];
    int unsafe = 0;

    for(int i=0;i<20;i++) cin >> v[i];

    float max = v[0], min = v[0];

    for(int i=0;i<20;i++) {
        if(v[i] > max) max = v[i];
        if(v[i] < min) min = v[i];

        if(v[i] < 3.0 || v[i] > 4.2)
            unsafe++;
    }

    cout << "Max: " << max << endl;
    cout << "Min: " << min << endl;

    if(unsafe > 0)
        cout << "Warning: Unsafe readings!\n";

    return 0;
}
```

Q9 – Smart Grid Load Distribution (2D Array + Functions)

Description: Stores power consumption (MW) for 4 regions over 7 days in a 2D array. Functions calculate total per region, identify the peak demand day, and compute overall average.

Source Code (C++):

```
#include <iostream>
using namespace std;

int main() {
    int arr[4][7];

    for(int i=0;i<4;i++)
        for(int j=0;j<7;j++)
            cin >> arr[i][j];

    // total per region
    for(int i=0;i<4;i++) {
        int sum=0;
        for(int j=0;j<7;j++)
            sum += arr[i][j];
        cout << "Region " << i << " Total: " << sum << endl;
    }

    // highest day
    int maxDay = 0, max = 0;

    for(int j=0;j<7;j++) {
        int sum=0;
        for(int i=0;i<4;i++)
```

```

        sum += arr[i][j];

        if(sum > max) {
            max = sum;
            maxDay = j;
        }
    }

    cout << "Highest demand day: " << maxDay << endl;

    return 0;
}

```

Q10 – Classroom Attendance System (2D Array + Functions)

Description: Stores attendance (1=Present, 0=Absent) for 5 students over 7 days in a 2D array. Functions calculate each student's total attendance, find the top attendee, and compute overall class attendance percentage.

Source Code (C++):

```

#include <iostream>
using namespace std;

int main() {
    int att[5][7];

    for(int i=0;i<5;i++)
        for(int j=0;j<7;j++)
            cin >> att[i][j];

    int total = 0;

    for(int i=0;i<5;i++) {
        int sum=0;
        for(int j=0;j<7;j++) {
            sum += att[i][j];
            total += att[i][j];
        }
        cout << "Student " << i << " attendance: " << sum << endl;
    }

    cout << "Class Attendance %: " << (total/(5.0*7))*100 << endl;

    return 0;
}

```